Do the weekly and monthly videos coach better when their prompts ask for a coach's moves?

The daily video's bullet, adapted and added to the weekly and monthly routines' prompts, replayed on ten past periods

Chess Coach research lab

Abstract

The app's weekly and monthly routines write a narrated video beside each period's review. The daily video's prompt took five moves of a strong human coach after a blind test: ask his move first, name the habit behind the mistake and the habit that replaces it, play the better move on to the opponent's best reply, say whether the miss is normal at his level, and praise a good move by what it saw. We added the same bullet, adapted to the period video, to the weekly and monthly prompts and replayed their review and video step on ten past periods (eight weeks and two months), once under each prompt, with the routines' own model and checker. Judging blind in both orders, a model preferred the candidate's coaching on all ten periods, train and held out alike, and its videos overall on seven periods against one. An audit against the evidence found the candidate asking his move first in 95% of its moment scenes (today's prompts: 16%), playing the better move on to their reply in 83% (28%) and placing the miss at his level in 95% of the positions it teaches (11%), with no more false claims, and the judge's listening verdicts were even. Every condition of the rule fixed before the first run held but one: heard once, the candidate's videos are 1.37 times as long as today's, past the bound of 1.3, so the rule keeps today's prompts. In the app's own voice the candidate's weekly videos last 7.0 minutes, inside the five to eight the video grammar asks for, where today's last 5.1. On the owner's call, both prompts now carry the bullet as tested (chess PR 167).

1 The question

The app's weekly and monthly routines have a model review the player's games of the period and write, beside the written review, a narrated video script, which the app renders as boards, charts, captions and a synthetic voice. A period video synthesizes: its grammar asks for two or three board scenes that show the period's recurring pattern through its clearest example, with how often it came back, among chart, tile and bullet scenes, and closes on the period's one focus.

The lab's daily study (*video-coaching*) tested five moves of a strong human coach in the daily video: ask before telling; name the habit behind the mistake, then the habit that replaces it; follow the better move to the opponent's best reply; say whether a miss is normal at the player's level; and praise a good move by what it saw. The daily prompt took them. The weekly and monthly prompts say how the video binds its boards and numbers but ask for none of the five. The candidate adds the daily's bullet to step 4 of both prompts, the step that writes the annotations and the video script, with three changes: it covers every board scene that teaches a move of his; the habit sub-point also says how often the habit came back in the period; and the engine's reply and the findability come from the day's archived analysis, since a period has no day's digest:

The video coaches the way a strong coach works with a player at his level, in every board scene that teaches a move of his, from the evidence alone and inside the period video grammar's chapters and rules:

- ask before you tell: the scene's first pause asks what he would play there, open ("What would you play here? Pick two moves."), before any question that narrows it to a kind of move and before any arrow;
- name the habit, then its replacement: say what kind of move he kept playing and how (its scan step, the seconds it got, what it left behind, as the dailies and the archive say) and how often it came back this period, and show at the board how the rule that answers it replaces that habit;
- and then what: after the better move, play or say their best reply and why the move still holds, so the line stops where its gain is clear (the turning point's engine line in coach-log/review-analysis/<date>.json, or `npm run --silent coach -- engine best <fen>`);
- pitch it at his level: say whether this is a miss most players at his rating make or one he could have found, from how often players at his band play the best move there (the `find` the day's Maia step archived on that turning point in coach-log/review-analysis/<date>.json, or `npm run --silent coach -- engine find <fen> <uci> --rapid <his rating>`), in words and never as a percentage (under about 15%: a pattern to learn; half or more: his to find); on a hung piece, the mirror instead, as the video grammar says;
- praise with a board: on a move he got right, name what it saw.

Nothing else in the prompts changes. The question is whether the period video coaches better with it, without becoming less true or much longer.

2 The cases and the writer

The cases are the ten periods with a review and a video script on the app's main branch from its command-line era: weeks 35 to 42 and the months of August and September 2026. Each is a checkout of the app at `3eb8afd0c1`, with the coach data as main held it just before that period's review landed, and the period's brief prepared again by the app's own `coach weekly prepare` or `coach monthly prepare`, offline. A week holds 19 to 59 games and the months 307 (August) and 172 (September). The split was fixed before any run, interleaved by date: train is weeks 35, 37, 39 and 41, and August; held out is weeks 36, 38, 40 and 42, and September.

The daily study's writer wrote the video alone, beside the review the day published. That does not work for a period: nine of the ten published reviews fail today's strict check against their prepared brief (22 to 271 errors), mostly for numbers the brief prepared again no longer carries and, in the older reviews, for sections the check now requires; only week 42's passes. So each run writes the whole of step 4, the review and its video script together, as the live routine does, and the graders read only the video script.

The writer runs as the routines do: Opus 5.5, headless, with nothing inherited from the session that launched it, and a short preamble that names the period, says the brief is ready and stops the writer once the strict check passes (no render, publish or commit). Each prompt wrote every period once, and today's prompts wrote weeks 35, 36, 37 and 38 a second time, to measure the judge's noise floor (24 runs in all).

3 The graders

A script with no model reads each run: whether the final check passes, the check rounds, commands it should not have run, the script's shape, its length heard once (spoken words at 2.6 a second plus every pause), cost and minutes.

Sonnet 5.5 audits each script alone against the evidence it was written from: the brief without its bulk, the period's turning points with his think times, a line per game, the period's and the career's digests, the week's daily logs or the month's weekly reviews, and for every board scene the position as the archive replays it from the day the board names, what was played there and next, the engine's best move and reply, the findability and the engine lines the day archived at that turning point, how players at his band play there (Maia-2), the plan lines, the threats, what each piece hits and guards, what each move the scene plays does, after each move the game did not play the engine's best reply, and the time he took on each of his moves in the scene's game with the clock he had left. It lists the claims the evidence contradicts (*false*), does not carry (*unsupported*) or carries but slants (*misleading*), and for each board scene its kind (a moment, praise, a judgement call, a plan, other), its false claims about the board, and which of the five moves it makes.

Sonnet 5.5 also judges each pair blind, in both orders, on four criteria: truth, coaching (with the period's own duty first: name what repeated and show it on the board), listening (would he follow it once, at speaking pace) and overall. A side wins a criterion only when both orders agree. Two controls check the judge on every period: today's script against itself, and against a decoy that adds as many spoken words as the candidate added on that period (at least 40), in beats that restate their scene with no square, move, piece, number or question the scene does not already have.

The pilot wrote week 35 under both prompts. Its candidate cited the long thinks that led to a turning point, which the evidence then could not check, so the evidence gained his time on every move of a scene's game, and every period, week 35 again included, was graded with it. The regrade's verdicts are the ones reported (Section 10).

4 The rule

A period scores $+1$ when the candidate wins a criterion, -1 when it loses and 0 for a tie; intervals are 95% percentile bootstrap intervals over periods. The candidate replaces the live prompts only when every condition of Table 1 holds, fixed before its first run.

Condition	Estimate (95% interval)	Holds
Coaching on the train periods above 0	+1.00 (+1.00 to +1.00)	yes
Coaching on the held-out periods above 0	+1.00 (+1.00 to +1.00)	yes
Overall not below 0	+0.60 (+0.20 to +1.00)	yes
False claims per script not up	−0.90 (−2.30 to +0.60)	yes
False board claims per script not up	0.00 (−1.00 to +1.00)	yes
The check passes as often	10 of 10 (today's: 10)	yes
Controls clean (noise floor: coaching)	0.00 (−0.75 to +0.75)	yes
Coaching moves made more often	+0.36 (+0.24 to +0.47)	yes
At most 1.3 times as long	1.37 (1.20 to 1.52)	**no**

Table 1: The rule, fixed before the candidate's first run. Coaching and overall are the judge's scores per period; the claim rows and the coaching moves are the audit's numbers per script, candidate minus today's prompts; the length is the candidate's script heard once over today's on the same period.

5 What the bullet changed

In the moment scenes	Published	Today's prompts	Candidate
Asks what he would play before any hint	15% (20)	16% (19)	95% (21)
Names the habit behind the mistake	90% (21)	84% (19)	76% (29)
Ends on the habit that replaces it	81% (21)	79% (19)	74% (31)
Plays out their best reply to the better move	19% (21)	28% (18)	83% (24)
Says whether the miss is normal at his level	38% (21)	11% (19)	65% (31)
and that agrees with the evidence	63% (8)	100% (2)	90% (20)
Counted once per position taught			
Names the habit behind the mistake	90% (21)	84% (19)	100% (21)
Ends on the habit that replaces it	81% (21)	79% (19)	100% (21)
Says whether the miss is normal at his level	38% (21)	11% (19)	95% (21)
Praise names what the good move saw	–	–	100% (11)
Any board scene: an open question first	30% (33)	38% (34)	98% (46)

Table 2: The coaching moves the audit finds, pooled over the ten periods: the share of the scenes that owe a move and make it, with the number owing it in brackets. A moment scene owes the habit, its replacement and the level; it owes the open question only when it asks him something, and the reply only when it shows a better move. In the middle rows, consecutive board scenes on the same position count once, and a move counts as made when any of them makes it.

Today's prompts name the habit behind a mistake and end on its replacement in most moment scenes, as the published scripts do, but they rarely ask his move before giving a hint (16%), rarely play the better move on to the opponent's best reply (28%) and rarely say whether a miss is normal at his level (11%). The candidate does all three in most moment scenes (95%, 83% and 65%), the level it states agrees with the findability evidence in 90% of the scenes that state one, and every praise scene it writes names what the move saw. Over the five moves, the share the audit finds in the moment scenes rose from 44% to 79% (+0.36 (+0.24 to +0.47) a script; rule 7). It also shows more boards (6.0 board scenes a script against 3.6) and teaches more moments in them (3.1 against 1.9).

Scene by scene, the candidate names the habit and its replacement a little less often (76% and 74%),

for the reason the daily study found: it often shows a position in two scenes, one that asks and one that answers (1.2 board scenes a script repeat the position of the board scene before them, against 0.0 under today's prompts), and the scene that asks counts as a miss. Counted once per position taught, it names the habit and ends on its replacement in 100% and 100% of them, against 84% and 79%.

6 What the judge preferred

Criterion	Train (five periods)	Held out (five periods)	All
Truth	+0.20 (−0.40 to +0.80)	−0.20 (−0.80 to +0.40)	0.00 (−0.50 to +0.50)
Coaching	+1.00 (+1.00 to +1.00)	+1.00 (+1.00 to +1.00)	+1.00 (+1.00 to +1.00)
Listening	−0.20 (−0.80 to +0.40)	0.00 (−0.80 to +0.80)	−0.10 (−0.60 to +0.40)
Overall	+0.60 (−0.20 to +1.00)	+0.60 (+0.20 to +1.00)	+0.60 (+0.20 to +1.00)

Table 3: The judge's mean score per period from the candidate's side (+1 a win, −1 a loss, 0 a tie), with its 95% bootstrap interval over periods.

The judge preferred the candidate's coaching on all ten periods, in both orders each time, and it preferred the candidate overall on seven periods against one, with two ties (Table 3). Truth was even: three wins, three losses and four ties. So was listening: three wins, four losses and three ties. The daily study's candidate never won listening; this one won it about as often as it lost it.

Two runs of today's prompts on the same week, judged against each other, split on coaching on two of four weeks, once each way, for a mean of 0.00 (−0.75 to +0.75). The judge does pick a side between two scripts of one prompt, but not one way on every period, as it did for the candidate. Read in one order only, the judge favoured the side it read second in 54% of its 102 decided verdicts, so a verdict that only one order gives is a tie.

7 The controls

Today's script judged against itself tied on every criterion on all ten periods. The decoy, today's script with on average 326 spoken words added that only restate their scene, never won coaching (it lost it on nine periods) and lost listening and overall on all ten. The judge does not reward words that say nothing new, so the candidate's wins come from what its extra words say.

8 Truth

The audit found fewer false claims in the candidate's scripts than in today's (1.40 and 2.30 a script; the difference is −0.90 (−2.30 to +0.60)), within the noise, and as many false claims about the board (0.90 against 0.90 a script; 0.00 (−1.00 to +1.00)), though the candidate shows more board scenes. It carries more lines the evidence does not settle (5.1 unsupported a script against 3.4) and about as many misleading ones (1.9 against 2.2).

9 Length, cost and the check

	Grammar	Published	Today's prompts			Candidate		
	minutes	heard once	heard once	rendered	inside	heard once	rendered	inside
Weeks	5 to 8	7.9	6.3	5.1	4 of 8	8.9	7.0	7 of 8
Months	6 to 9	6.6	6.5	5.1	0 of 2	7.5	5.8	1 of 2

Table 4: Minutes per video: heard once (spoken words at 2.6 a second plus every pause, the rule's measure) and rendered (the app's plan voiced by its own voice, the measure the video grammar's minutes use), and how many rendered videos fall inside the grammar's minutes.

Every run of both prompts passed the routine's check. Heard once, the candidate's videos last 8.6 minutes against 6.4, with 1165 spoken words against 873 and 9.7 questions against 4.2; on the same period it is 1.37 (1.20 to 1.52) times as long, past the 1.3 rule 8 allows. Only on August was the candidate's video the shorter.

The rule measures length heard once, at a narrator's pace. The video grammar gives its minutes for the rendered video, and the app's voice is quicker: rendered, a video lasts on average 79% of its length heard once (the app's own estimate of it, from the words alone, is within 6% of the voice on every script). Rendered, the candidate's weekly videos last 7.0 minutes and today's 5.1 (Table 4); on the same period the candidate is 1.35 times as long rendered, about as much as heard once. The grammar asks for five to eight minutes a week: seven of the candidate's eight weeks fall inside it and one runs past it, against four of today's, whose other four run short. The months are two and say little: rendered, the candidate's August and September videos last 4.4 and 7.3 minutes and today's 5.8 and 4.4, against the grammar's six to nine. This comparison was made after the result and decides nothing.

The writer took 7.1 check rounds against 6.3, 19 minutes against 18 and $6.48 against $6.13 a period at list price.

10 The graders measured twice

The pilot's four scripts (week 35 under both prompts and the published scripts of week 35 and August) were audited with the first evidence and again with the final one. Their coaching share moved by 10 points on average and by up to 26, so one period's share in Table 5 says little, while the pooled difference (+0.36 (+0.24 to +0.47)) is well beyond that. The judge's verdicts on week 35 were the same on every criterion both times.

			Judge, from the candidate's side				Minutes heard once		Coaching share	
Period	Split	Games	Truth	Coaching	Listening	Overall	Today	Candidate	Today	Candidate
Week 35	train	55	+	+	−	+	6.5	9.5	40	73
Week 37	train	19	0	+	−	+	6.1	9.3	50	100
Week 39	train	59	−	+	0	−	6.2	7.7	50	69
Week 41	train	34	0	+	0	+	6.5	10.4	60	69
August	train	307	+	+	+	+	7.3	5.6	44	56
Week 36	held out	32	0	+	0	0	5.6	8.2	40	59
Week 38	held out	49	−	+	−	0	6.1	9.3	40	93
Week 40	held out	33	+	+	+	+	6.7	8.3	40	100
Week 42	held out	34	−	+	+	+	7.0	8.5	40	100
September	held out	172	0	+	−	+	5.6	9.4	30	71

Table 5: Each period: the judge's verdict from the candidate's side (+ a win, − a loss, 0 a tie), the minutes heard once under each prompt, and the share of the coaching moves the audit finds in the moment scenes.

11 What the app takes

The rule keeps today's prompts: the candidate held every condition but the length (Table 1). It won the coaching criterion on every period, with no more false claims and even listening verdicts, and its extra length comes with the three moves today's prompts skip: a question before each answer, the reply played out, and the level. Whether that length is worth it was the owner's call, which the rule, fixed before the first run, does not make. The owner took it: the weekly and monthly prompts carry the bullet as tested (chess PR 167). A next round could ask for the same moves in fewer words, as the daily study's next round would.

12 Limits

- The five moves are one change: the result shows they help together, not which of them does.
- Each run writes the period's review and its video together, so two runs of a period may show different boards; the judge compares two videos, not two treatments of one board.
- A replay writes the review and the video after the brief is in place; the routine writes them at the end of a longer run. Over the last four weeks (weeks 39, 40, 41 and 42) the published videos ran about as long heard once as today's replays of the same weeks (6.5 against 6.6 minutes), so the replays' lengths look like the routine's; the earlier published weeks ran longer.
- The graders are models. The judge picks a side between two runs of one prompt on some periods and favours the side it reads second; the audit's tallies move between two readings. The controls, the noise floor and both orders bound these, and the graders are a different model from the writer.
- Ten periods, two of them months: the months' results stand on one period each.
- The level a script states is checked against the same findability the writer reads (Maia-2 at his band), so the audit checks that the writer read it right, not that Maia-2 knows his level.
- The eval spent $225 at list price ($155 on the writer, $63 on the graders, $6.50 on runs a container restart cut short), within its cap of $276, on the owner's subscription, whose seven-day window it moved from 55% to 68%.